

Scalability & Future Plan

Date	30 September 2026
Team ID	SWTID-2026-9716
Project Name	Weather Wise
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Weather data depends on the availability of the weather API	Weather information may not load when the API is unavailable	High
2	Application performance depends on internet connectivity	Slow internet may cause delays in fetching weather data	Medium
3	Limited weather features in the current version	Users may need additional information and alerts	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed for a limited number of users	Use scalable cloud hosting and server resources
2	Data Storage	Limited storage requirements	Use a scalable cloud database if historical weather data is added
3	Performance	Weather data is fetched through an API	Use caching and optimized API requests to improve response time
4	Security	Basic API key protection	Use secure environment variables and authentication where required

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Add detailed multi-day weather forecasts	1–2 Months	Helps users plan activities in advance
Phase 3	Add weather alerts and notifications	2–3 Months	Provides timely information about important weather conditions
Phase 4	Add historical weather data and weather analytics	3–6 Months	Helps users understand weather patterns and trends